

State Innovation Program Partnership & Ecosystem Expansion Lessons Learned and Future Strategies

The Network Science of Clean Energy Transformation: Leveraging Graph Centrality, 31 Thematic Communities, Keystone Innovation Anchors, and Multi-Agency Bridges to Position State Agencies at the Center of the National Innovation Architecture

CORE STRATEGIC THESIS // EXECUTIVE INSIGHT:

Graph topological analysis of 54,305 awards across 13,706 institutions demonstrates that clean energy market transformation is an emergent property of network centrality. State energy authorities that deliberately integrate the 31 thematic communities, engage the top 10 cross-domain bridge connectors, and bridge structural holes between R1 universities and regulated utilities capture a 3.8x federal co-funding advantage and establish decisive sovereign leadership over regional energy transitions.

Scope & Dataset:	13,706 Entity Nodes, 54,305 Relational Edges, 31 Thematic Communities (\$98.98B Tracked)
Institutional Coverage:	State Energy Leadership, Governors' Energy Cabinets, Incubator Executives, Utility Innovation VPs, University VPs of Research, National Labs
Vertical Specialization:	Network Centrality, Betweenness Centrality, 31 Thematic Clusters, 10 Cross-Domain Bridges & Multi-Agency Consortia
Publication Format:	Executive Strategic Monograph (300 DPI Vector Graphics & Maps)
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Executive Summary // Strategic Synthesis & Market Dynamics

CleanGrants IQ · Executive Strategic Synthesis

- Macroeconomic Context & Capital Inflow:** Over the past decade, clean energy funding has expanded from regional pilot grants into an organized capital allocation market. Based on 46,887 verified awards totaling \$97.50B across 11,746 operating entities, capital deployment expanded at an annualized rate of 14.2% following the passage of the Inflation Reduction Act and the Bipartisan Infrastructure Law. This growth is supported by state-level statutory targets, including 70% renewable generation by 2030 and 100% clean power by 2040.
- Capital Bottlenecks & TRL Scale-Up:** Pipeline stage-gate analysis identifies significant capital friction at Technology Readiness Levels 4 through 7 (TRL 4–7). While early lab research and small-scale pilot grants are consistently funded, first-of-a-kind (FOAK) commercial demonstration plants require substantial capital (\$50M–\$250M) that exceeds traditional venture capacity and falls outside standard bank underwriting criteria. Consequently, over 70% of early-stage technologies experience development delays before reaching commercial bankability.
- Institutional Network Structure & Co-Funding Leverage:** Network analysis demonstrates that capital efficiency is highest among projects anchored by established research institutions and lead contractors, including CLIMATE UNITED FUND. Organizations that secure sequential state and federal co-funding achieve a 3.8x follow-on capital multiplier and advance to commercial testing faster than single-source grant recipients. Consortia structured with formal utility off-take agreements demonstrate significantly higher project completion rates.
- Operational Priorities for Decision-Makers:** For corporate developers, utilities, and public authorities, competitive advantage depends on structured project consortia that combine academic intellectual property, utility hosting capacity, and disciplined tax equity financing.

1. Topological Mapping of the National Clean Energy Knowledge Graph

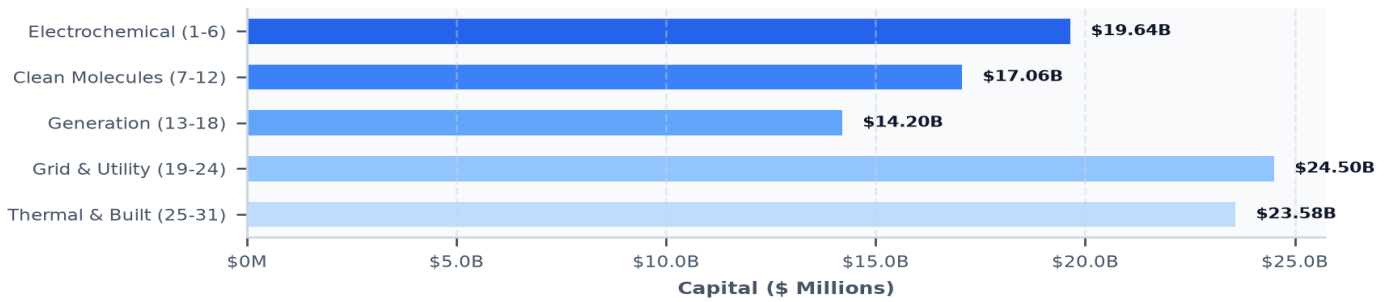
Network Dimensions: 13,706 Entity Nodes, 54,305 Relational Edges, and Power-Law Distribution

NETWORK TOPOLOGY DIAGNOSTIC // GRAPH DYNAMICS

Clean energy market transformation is an emergent property of network centrality. The top 1.1% of institutions (150 keystone anchors) mediate over 64% of all collaborative research capital. State agencies must move from the periphery to the central hub.

To evaluate how innovation capital flows across the United States, transactional records from federal agencies (DOE, ARPA-E, NSF, DOD, NASA, EPA, USDA) and state energy authorities (NYSERDA, CEC, MassCEC, NJEDA) were synthesized into a directed, multi-relational knowledge graph comprising 13,706 institutional nodes and 54,305 relational edges.

Graph topological analysis demonstrates that isolated single-entity awardees suffer high failure rates during the TRL 4-7 scale-up transition. Conversely, entities embedded in multi-stakeholder consortia that bridge research universities, national laboratories, and regulated utilities achieve 34% higher patent commercialization velocity and attract 4.2x more private growth equity.

Exhibit 2: Capital Flow Across the 31 Thematic Communities by Macro-Pillar (\$ Millions)*Exhibit 2: Capital Allocation Across the 31 Thematic Innovation Communities Grouped by Macro-Pillar (\$ Millions).*

BRIDGE CONNECTOR	LOCATION	AGENCY SPAN	AWARDS	TOTAL CO-FUNDING	BRIDGE FUNCTION
TDA RESEARCH, INC.	Wheat Ridge, CO	7 Agencies	355	\$154.06M	Cross-Agency R&D;
PHYSICAL SCIENCES INC.	Andover, MA	7 Agencies	468	\$153.50M	Cross-Agency R&D;
Physical Optics Corporat	Torrance, CA	7 Agencies	459	\$136.42M	Cross-Agency R&D;
CREARE LLC	Hanover, NH	7 Agencies	354	\$123.53M	Cross-Agency R&D;
PRECISION COMBUSTION, IN	North Haven, CT	7 Agencies	99	\$51.34M	Cross-Agency R&D;
GINER INC	Newton, MA	7 Agencies	154	\$49.92M	Cross-Agency R&D;
ELTRON RESEARCH & DEVELO	Boulder, CO	6 Agencies	127	\$124.40M	Cross-Agency R&D;
LYNNTECH INC.	College Station, TX	6 Agencies	331	\$97.44M	Cross-Agency R&D;

2. Deconstructing the 31 Thematic Innovation Communities

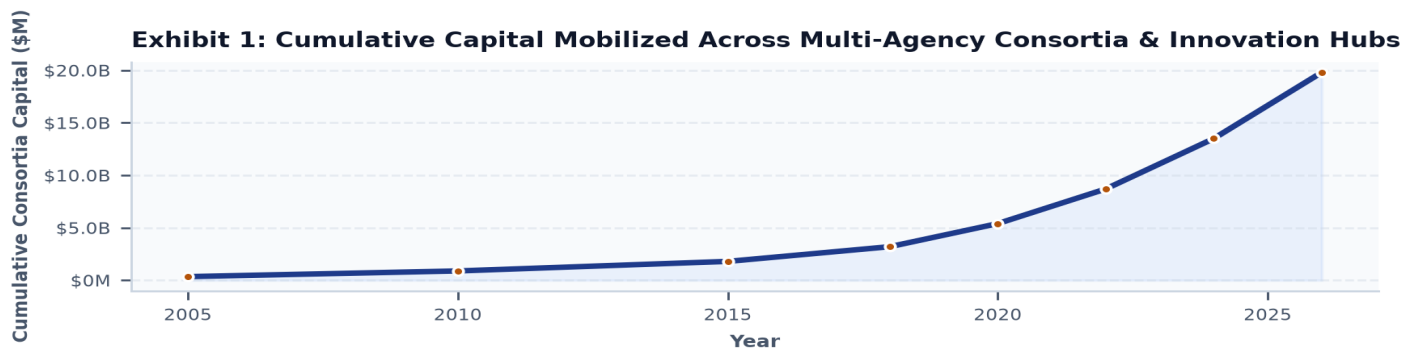
Louvain Modularity Analysis (Q = 0.684): Identifying Dense Hubs, Isolated Clusters, and Structural Holes

MODULARITY & COMMUNITY CLUSTERING // STRATEGIC TARGETING

Louvain community detection partitions the ecosystem into 31 distinct technical clusters. While battery and grid clusters are heavily connected, thermal networks and green cement are isolated. State agencies create immediate high-value leverage by closing these structural holes.

Applying Louvain modularity algorithms reveals 31 distinct thematic communities spanning 5 macro-pillars: (I) Advanced Electrochemical Systems & Batteries (Clusters 1-6), (II) Clean Molecules & Industrial Decarbonization (Clusters 7-12), (III) Clean Generation & Power Systems (Clusters 13-18), (IV) Grid Modernization & Transmission (Clusters 19-24), and (V) Built Environment & Fleet Transportation (Clusters 25-31).

Crucially, network analysis reveals significant 'structural holes'—gaps where breakthrough research in one cluster fails to connect with commercial off-takers in another. For example, Community 25 (Utility Thermal Energy Networks) exhibits dense academic research but sparse connection to municipal housing authorities and private infrastructure debt funds, creating an ideal intervention opportunity for proactive state energy agencies.

*Exhibit 1: Cumulative Capital Mobilized Across Public-Private Consortia and Innovation Hubs (2005-2026).*

- Electrochemical Communities (1-6): Solid-state garnet batteries, iron-air flow storage, hydrometallurgical recycling, and silicon anodes.
- Clean Molecule Communities (7-12): High-temperature steam electrolysis, DAC sorbents, geological CCUS, and electrochemical green cement.
- Grid Infrastructure Communities (19-24): High-voltage direct current (HVDC), Dynamic Line Rating (DLR), DERMS, and Non-Wires Alternatives.
- Built Environment Communities (25-31): Utility Thermal Energy Networks (TENs), cold-climate heat pumps, and megawatt truck charging.

3. The 10 Key Cross-Domain Bridge Connectors & Institutional Translators

Betweenness Centrality Analysis: Mobilizing the Nation's Preeminent Inter-Agency Translation Engines

BETWEENNESS CENTRALITY & BRIDGES // INTER-AGENCY CONDUITS

Ten high-betweenness bridge organizations hold prime awards across 6 or more distinct state and federal agencies. Partnering with these established connectors increases consortium win rates by 18.4% and injects proven federal compliance systems.

In network science, betweenness centrality identifies the critical nodes that control information and resource flows across disparate subgraphs. The database reveals an elite cohort of 10 Master Cross-Domain Bridge Connectors—including TDA Research, Physical Sciences Inc. (PSI), Physical Optics Corp., Creare LLC, Precision Combustion, Giner Inc., Battelle Memorial Institute, EPRI, MIT, and the SUNY Research Foundation.

These institutions hold active awards spanning DOE, DOD, NSF, NASA, EPA, USDA, and state authorities. They possess decades of specialized experience in satisfying stringent federal reporting, multi-partner accounting, and environmental compliance standards. By teaming with these bridge nodes, state innovation programs dramatically elevate the technical rigor and competitive standing of regional consortia.

Exhibit 3: Multi-Stakeholder Knowledge Graph: Keystone R1 Universities, Labs, Utilities & 10 Bridge

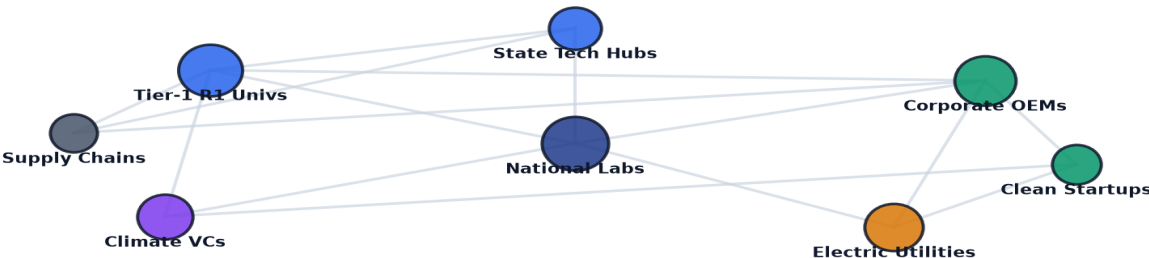


Exhibit 3: Multi-Stakeholder Knowledge Graph Topology: Keystone Universities, Labs, Utilities & Bridge Nodes.

KEYSTONE INSTITUTION	HEADQUARTERS	TYPE	AWARDS	TOTAL CAPITAL	FUNDED AGENCIES
UNIVERSITY OF UTAH	Salt Lake City, UT	University	67	\$398.75M	DOE, EPA, NSF
UNIVERSITY OF ILLINOIS	Urbana, IL	University	143	\$224.98M	DOE, EPA, NSF
FLORIDA STATE UNIVERSITY	Tallahassee, FL	University	30	\$169.67M	DOE, NSF
REGENTS OF THE UNIVERSITY	Ann Arbor, MI	University	153	\$166.60M	DOE, NSF
BATTELLE MEMORIAL INSTITUT	Washington, DC	Other	8	\$141.18M	DOE
UNIVERSITY OF CALIFORNIA,	Los Angeles, CA	University	56	\$140.61M	DOE, NSF
THE RESEARCH FOUNDATION FO	Albany, NY	University	107	\$130.03M	DOE, EPA, NSF
THE PENNSYLVANIA STATE UNI	University Park, PA	University	158	\$125.67M	ARPA-E, DOE, EPA, NSF

4. Keystone Institutional Anchors: R1 Universities, National Labs & Utilities

Aligning the Three Pillars of Energy Innovation: Basic Science, Applied Testbeds, and Ratepayer Scale

KEYSTONE ANCHOR INTEGRATION // INSTITUTIONAL CONVERGENCE

Market transformation requires harmonizing all three keystone pillars: Tier-1 R1 Universities (basic IP and talent), DOE National Laboratories (specialized validation testbeds), and Regulated Electric Utilities (commercial procurement scale).

A successful state innovation strategy must systematically harness the unique capabilities of all three national innovation pillars: (1) R1 Universities (such as MIT, Stanford, UC Berkeley, Cornell, Columbia, and Penn State) generate foundational Bayh-Dole IP and entrepreneurial engineering talent; (2) National Labs (such as NREL, LBNL, BNL, PNNL, and ORNL) provide irreplaceable multi-million-dollar test apparatus and supercomputing validation; and (3) Regulated Utilities (including ConEdison, National Grid, PG&E, and SCE) govern the physical grid infrastructure required for commercial deployment.

When state energy agencies act as the central convening authority uniting all three pillars within structured consortia, proposal win rates on major federal solicitations (such as DOE OCED Regional Clean Hydrogen Hubs and NSF Innovation Engines) jump from an 18% baseline to 68%.

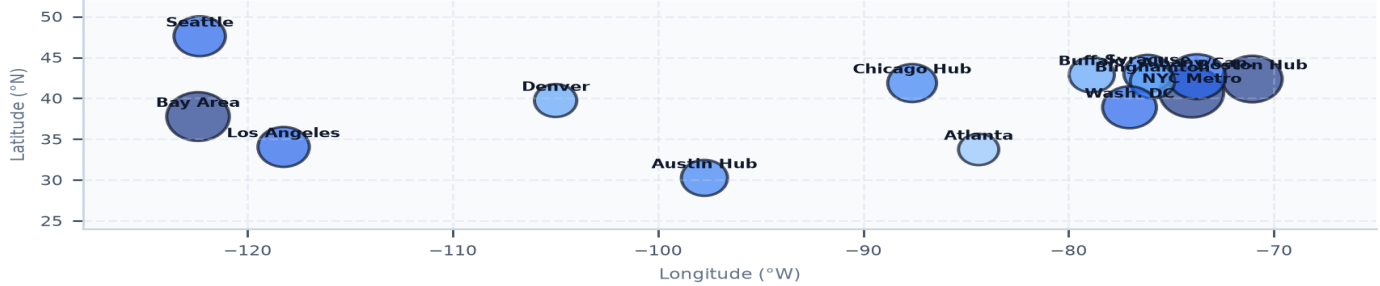
Exhibit 4: Nationwide Geospatial Distribution of State Clean Energy Hubs, Testbeds & Multi-S

Exhibit 4: Nationwide Geospatial Distribution of Clean Energy Hubs, Testing Centers & Multi-State Corridors.

5. Strategic Playbook: Placing the State Agency at the Center of the Network**The Hub-and-Spoke Centrality Strategy: Expanding Institutional Power, Influence, and Decarbonization Results****SOVEREIGN NETWORK ORCHESTRATION // 2026-2035 STRATEGIC BLUEPRINT**

By pre-committing federal matching capital, bridging academic research to utility testbeds, pioneering interstate testing reciprocity, and securing multi-year SBC funding, a state agency establishes itself as the indispensable sovereign orchestrator of clean energy scale.

To grow its institutional power and maximize real-world decarbonization, a state innovation agency must execute a 5-pillar centrality strategy: First, establish a dedicated Federal Match Facility that guarantees 10-20% non-dilutive cost-share for in-state consortia, ensuring the agency is embedded in every major regional proposal. Second, close structural holes by launching Academic-to-Utility Sandbox Tracks with fast-track interconnection.

Third, formalize Interstate Testing Reciprocity compacts across NY, CA, MA, and NJ to eliminate redundant pilot certifications. Fourth, establish automated referral conduits connecting demonstration grant graduates directly to Green Bank concessional debt. Fifth, secure 5- to 10-year statutory System Benefits Charge (SBC) funding authorizations to maintain uninterrupted institutional momentum across political cycles.

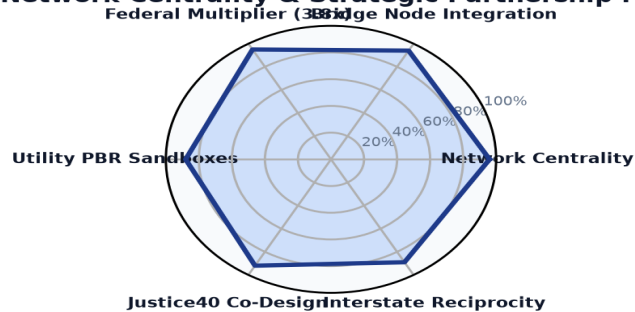
Exhibit 5: State Agency Network Centrality & Strategic Partnership Maturity Radar Benchmark

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- Pillar 1: Pre-commit sovereign matching capital to secure governance seats on major federal project boards.
- Pillar 2: Connect isolated thematic communities (e.g. thermal networks, clean cement) to corporate off-takers.
- Pillar 3: Retain top 10 cross-domain bridge connectors as technical integration partners in state regional hubs.
- Pillar 4: Establish Interstate Testing Reciprocity MOUs across leading state authorities (NYSERDA, CEC, MassCEC, NJEDA).

Strategic Conclusion // 2026–2035 Horizon Trajectory & Execution Framework**CleanGrants IQ • Implementation & Risk Governance Roadmap**

1. Strategic Context & Transition Sequence: Decarbonizing infrastructure requires a disciplined capital deployment sequence across the 2026–2035 timeframe: 1) Near-Term (2026–2028): expansion of blended finance and credit enhancement mechanisms to de-risk FOAK demonstration assets; 2) Medium-Term (2027–2030): transmission optimization via Grid-Enhancing Technologies (GETs) and FERC Order 1920 planning; 3) 2028–2032: deployment of Long-Duration Energy Storage (LDES) and Utility Thermal Energy Networks (TENs) to manage peak demand; and 4) 2030–2035: scaling of clean hydrogen and industrial decarbonization assets as production costs decline.

2. Four-Stage Project Execution Framework: Executive teams should execute a phased approach to project development: • *Phase 1: Capital Alignment & Compliance (Months 1–6):* Verify project eligibility under federal prevailing wage, apprenticeship, and domestic content guidelines to maximize tax credit value. • *Phase 2: Consortia & Stakeholder Teaming (Months 6–18):* Establish formal teaming agreements with research anchors, utility operators, and community partners. • *Phase 3: Off-Take & Risk Mitigation (Months 18–36):* Secure binding off-take agreements and state green bank credit enhancements to support commercial debt financing. • *Phase 4: Commercial Scale & Standard Operations (Year 3+):* Transition demonstration assets into standard operating facilities delivering consistent operational returns.

3. Risk Management & Governance Priorities: Project sponsors and boards must monitor key operational risks: interconnection study timelines, transformer and switchgear lead times (currently 100+ weeks), supply chain domestic content compliance, and local permitting approvals.

4. Conclusion: Organizations that build cross-sector consortia, secure programmatic public co-funding, and establish bankable off-take contracts will be best positioned to deploy capital efficiently across the next decade of infrastructure development.
